9.7 Binomial Distributions

Name _____ Period ____ Date _____

Algebra I 9.7 (30-minute plan) · Student sheet

I can…

- make and read a probability distribution.
- tell when an experiment is binomial.
- find binomial probabilities and interpret a binomial distribution.

Vocabulary · fill in the missing word

Random variable — a variable whose value is _____ by the outcome of an experiment

Probability distribution — a table, graph, or rule that gives the probability of each value of a random _____; the probabilities add to 1

Binomial experiment — n _____ trials, each with two outcomes (success or failure), and the same probability p of success each time

Binomial probability — P(k _____) = $_nC_k \cdot p^k \cdot (1 - p)^{n-k}$

Part 1 · Notes and examples (fill in each note, then work the example)

Probability distribution: In a probability distribution, the probabilities add to _____.

Example 1

A bag has 3 balls labeled 1, 2 balls labeled 2, and 1 ball labeled 3. Let x be the number on a ball chosen at random. Make the probability distribution.

Binomial experiment: A **binomial experiment** has n _____ trials, each with _____ outcomes and the same probability p.

Binomial probability: P(k successes) = _____

Fair coin: For a fair coin: P(k heads in n flips) = _____ ÷ _____

Example 2

You flip a coin 10 times. What is the probability of getting exactly 7 heads?

Part 2 · Partner practice

Partner practice

You flip 6 coins. What is the probability of getting exactly 2 heads?